

Q1: $P(\text{Pass}) = 0.9$

$$P(\text{Quick} | \text{Pass}) = 0.6$$

$$P(\text{Quick} | \text{Not Pass}) = 0.3$$

$$P(\text{Not Pass}) = 1 - P(\text{Pass}) = 0.1$$

$$P(\text{Pass} | \text{Quick}) = \frac{P(\text{Pass}) \cdot P(\text{Quick} | \text{Pass})}{P(\text{Quick})}$$

$$\begin{aligned} P(\text{Quick}) &= P(\text{Pass})P(\text{Quick} | \text{Pass}) \\ &\quad + P(\text{Not Pass})P(\text{Quick} | \text{Not Pass}) \\ &= 0.9 \times 0.6 + 0.1 \times 0.3 \\ &= 0.57 \end{aligned}$$

$$\begin{aligned} \Rightarrow P(\text{Pass} | \text{Quick}) &= \frac{P(\text{Pass}) \cdot P(\text{Quick} | \text{Pass})}{P(\text{Quick})} \\ &= \frac{0.9 \times 0.6}{0.57} \\ &= 0.9474 \end{aligned}$$

Q 2 :

Likelihood

$$p(x|\theta, n) = \text{Multinomial}(n, \theta)$$

Prior

$$p(\theta|\alpha) = \text{Dirichlet}(\alpha)$$

Posterior

$$p(\theta|x, n, \alpha) = \frac{p(x|\theta)p(\theta)}{p(x)} = \int p(x|\theta)p(\theta)d\theta$$

= Dirichlet distribution

with $\alpha' = \alpha + x$

where $x = (x_1, x_2, \dots, x_k)$

The posterior is always proportional to the joint

$$p(\theta|x) = (1/c) \times p(\theta, x) \propto p(\theta, x)$$

and the posterior is always a proper distribution
if the prior is a proper distribution